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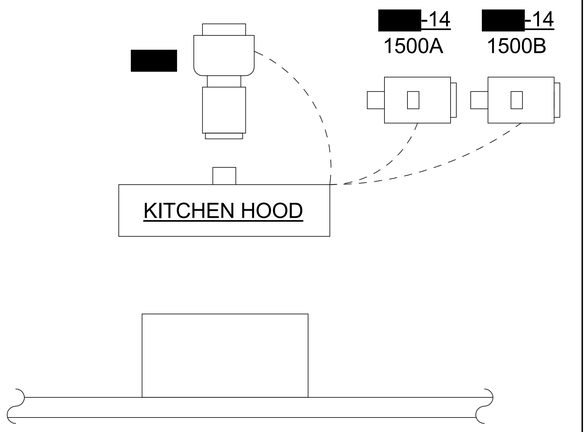
DEMAND CONTROL VENTILATION WITH CENTRALIZED CO2 SENSOR

PROJECT [redacted]	POINT NAME	BI	BO	BV	AI	AO	AV	ALARM TYPE	GRAPHIC
[redacted]UPIED SCHEDULE	[redacted]_SCHED						X		YES
UN[redacted]UPIED SCHEDULE	UN[redacted]_SCHED						X		YES
[redacted] SET POINT MINIMUM [redacted]UPIED	[redacted]_SP_MIN_[redacted]					X			YES
[redacted] SET POINT MAXIMUM [redacted]UPIED	[redacted]_SP_MAX_[redacted]					X			YES
[redacted] SET POINT STANDBY [redacted]UPIED	[redacted]_SP_STBY_[redacted]					X			YES
[redacted] SET POINT UN[redacted]UPIED	[redacted]_SP_UN[redacted]					X			YES
[redacted] SENSOR [redacted]UPIED	[redacted]_S_[redacted]	X							
[redacted] SENSOR UN[redacted]UPIED	[redacted]_S_UN[redacted]	X							
[redacted] ACTUAL	[redacted]_ACTUAL				X				YES
[redacted] DAMPER %OPEN	[redacted]_DAM_%				X				YES
ZONE CO2 LEVEL [redacted]	CO2_[redacted]				X				YES
ZONE CO2 SET POINT	CO2_SP					X			YES
CHANGE CO2 SENSOR ALARM	CO2_S_ALARM	X						[redacted]	YES



KITCHEN HOOD & EXHAUST CONTROL SEQUENCE

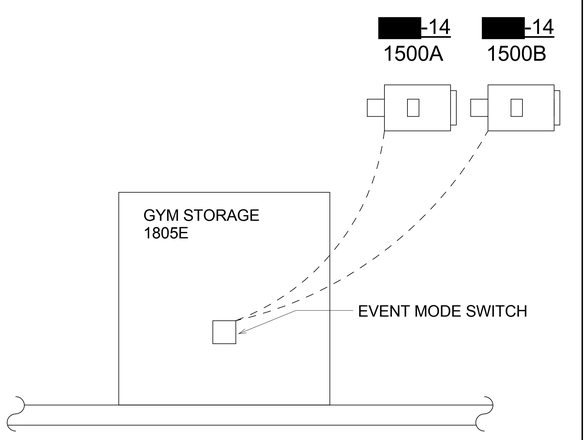
NOT TO SCALE

KITCHEN HOOD & EXHAUST CONTROL SEQUENCES:

1. The kitchen exhaust hood shall be interlocked with [redacted] 14 1500A and [redacted] 14 1500B, and normally open [redacted] motorized exhaust damper serving louver L-2.
2. Upon [redacted] of [redacted] 14 1500A and 1500B shall both be activated and [redacted]ch set to a minimum of 1500 [redacted]. Both boxes shall return to normal operation upon the [redacted] of exhaust [redacted].
3. Upon [redacted] of [redacted] Normally open [redacted] motorized exhaust damper shall be energized closed so that gym over pressurizes and transfers [redacted] for make-up into the kitchen. Exhaust damper shall return to normal position upon [redacted] of [redacted].

KITCHEN HOOD & EXHAUST CONTROL POINTS

POINT [redacted]	POINT NAME	BI	BO	BV	AI	AO	AV	ALARM TYPE	GRAPHICS
KITCHEN HOOD STATUS	DW_S	X							X
EXHAUST [redacted] COMMAND	DW_EF_C		X						X
[redacted] BOX COMMAND	DW_[redacted]_C		X						X
MOTORIZED EXHAUST DAMPER	[redacted]_D_C		X						X
MOTORIZED [redacted]XHAUST DAMPER STATUS	[redacted]_D_S	X							X

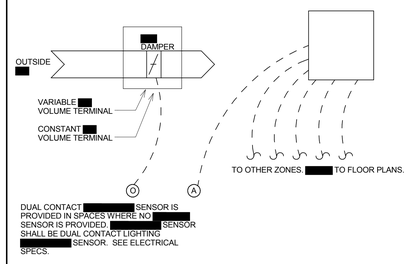


NOT TO SCALE

[redacted] OUTSIDE [redacted] CONTROL SEQUENCES:

1. [redacted] 1800 **normal mode** of operation shall be such that outside [redacted] is provided for classroom use. [redacted] 1800 normal minimum shall be 1200 [redacted] and normal maximum shall 1600 [redacted] box A serving gym shall maintain [redacted] values and modulate accord[redacted]g to CO2.
2. [redacted] 1600 **event mode** shall be initiated by a push button in Gym Storage 1805E and start a 2 hour timer ([redacted]) for an event mode which will automatically [redacted]able upon expiration of timer. [redacted] 1800 Event mode shall have a minimum [redacted] value of 1600 [redacted] and maximum value of 3000 [redacted] box A and B shall work together and modulate based on CO2 to maintain [redacted] values.
3. Note that Kitchen hood operation shall overide both of the above modes and take controls of [redacted] boxes A and B serving gym.

ALL SPACES SERVED BY VARIABLE [redacted] VOLUME TERMINAL OR CONSTANT [redacted] VOLUME TERMINAL



DEMAND CONTROL VENTILATION WITH CENTRALIZEDCO2 SENSOR([redacted])

1. DEMAND CONTROL VENTILATION SEQUENCE OF OPERATION [redacted]

1.0 [redacted]UPIED MODE:

1.1. [redacted] shall monitor CO2 levels in [redacted]ch zone.

1.2. [redacted] shall compare the CO2 levels in [redacted]ch zone with [redacted]. A [redacted]ferential CO2 signal shall be sent to the [redacted]. The signal shall be a 0-10 [redacted] signal where 0 [redacted] represents 200 [redacted]ferential, 5 [redacted] represents 500 [redacted] and 10 [redacted] represents 800 [redacted] [redacted]ferential.

1.3. The [redacted] box controller shall modulate the [redacted] box damper to maintain as follows: 200 [redacted] ([redacted]) damper is closed, 500 [redacted] damper is half open ([redacted]), and 800 [redacted] ([redacted]) damper is full open.

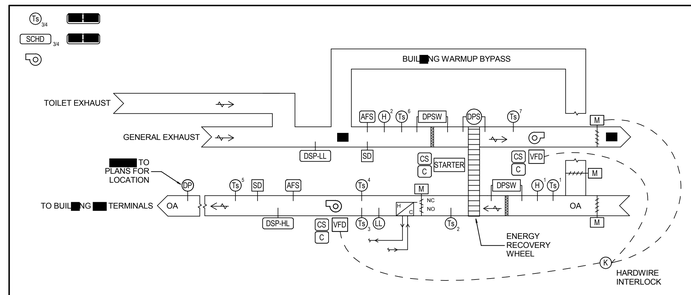
1.4. Controls contractor is responsible for prov[redacted]g field mounted [redacted] box controller to accomplish sequence and achieve all listed control points.

1.5 If a pace is not provided with an [redacted] sensor, [redacted] box shall respond to [redacted]upancy sensor. During [redacted]pied hours the [redacted] box shall have an [redacted]pied standby [redacted] setting that will enable when NO [redacted] is detected and an [redacted]upied [redacted] setting that will enable when [redacted] is detected. If [redacted] box serves more than one space then all space [redacted]upany sensors must detect NO [redacted] before [redacted] box is allowed to enable [redacted]pied standby [redacted] setting.

MECHANICAL CONTROL LEGEND

AFF	ABOVE FINISHED FLOOR
BAS	BUIL[redacted]NG AUTOMATION SYSTEM
CO_2	CARBON [redacted]OXIDE
TCC	TEMPERATURE CONTROL CONTRACTOR
DP	[redacted]
[redacted]	EXHAUST [redacted] PATH
RA	RETURN [redacted] PATH
SA	SUPPLY [redacted] PATH
[redacted]R	H[redacted]T PUMP WATER SUPPLY/RETURN
NC	NORMALLY CLOSED
OA	OUTSIDE [redacted] PATH
[redacted]	[redacted]
PRESS	PRESSURE
[redacted]	[redacted] INPUT
DO	[redacted] OUTPUT
AI	ANALOG INPUT
AO	ANALOG OUTPUT
VFD	VARIABLE FREQUENCY DRIVE
RH	RELATIVE HUM[redacted]TY
[redacted]	MAKE-UP [redacted] UNIT
O/H	[redacted]UPIED H[redacted]TING SETPOINT
O/C	[redacted]UPIED COOLING SETPOINT
[redacted]	[redacted] SETPOINT
[redacted]	[redacted] COOLING SETPOINT
(K)	EMERGENCY HVAC/VENTILATION KILL BUTTON
(■)	AVERAGING TEMPERATURE SENSOR

(Ts)	
(H)	
(LL)	
(P)	
(DP)	
DPSW	
DPS	
(C)	
(M)	
(F)	
CS	
SD	AVERAGING TEMPERATURE SENSOR
COS	AVERAGING TEMPERATURE SENSOR
DSP-HL	AVERAGING TEMPERATURE SENSOR
DSP-LL	AVERAGING TEMPERATURE SENSOR
ZN-DP	AVERAGING TEMPERATURE SENSOR
ZN-CO2	AVERAGING TEMPERATURE SENSOR
ZN-[redacted]	AVERAGING TEMPERATURE SENSOR
ZN-T	AVERAGING TEMPERATURE SENSOR
CO2	AVERAGING TEMPERATURE SENSOR
(O)	CENTRIFUGAL [redacted]



OUTSIDE [redacted] SYSTEM (OA-1) SCHEMATIC

NOT TO SCALE

OUTSIDE [redacted] SYSTEMS:

1. THERE ARE TWO OUTSIDE [redacted] UNITS IN THIS [redacted]
2. [redacted]CH SYSTEM SHALL OPERATE UNDER THE CONTROL OF A [redacted] STAND-ALONE MICROPROCESSOR BASED DDC CONTROLLER.
3. [redacted]CH SYSTEM SHALL BE PLACED INTO THE [redacted]UPIED/UN[redacted]UPIED MODE BASED UPON THE USER ADJUSTABLE SCHEDULE AT THE [redacted] CONTROLLER. THESE SYSTEMS SHALL BE IN THE [redacted]UPIED MODE DURING REGULAR SCHOOL HOURS ONLY. SEE [redacted]
4. IF COMMUNICATION IS LOST BETWEEN THE [redacted] CONTROLLER AND THE OUTSIDE [redacted] SYSTEM CONTROLLER, THEN THE OUSIDE [redacted] SYSTEM SHALL BE PLACED INTO THE UN[redacted]UPIED MODE UNTIL COMMUNICATION IS RESTORED.
5. [redacted]CH SYSTEM WILL BE PLACED INTO A MODE OF OPERATION BASED UPON THE FOLLOWING ADJUSTABLE TERMPERATURE SCHEDULE:

OUTSIDE [redacted] TEMPERATURE	MODE OF OPERATION
75 DEG F OR GR[redacted]TER ([redacted])	COOLING MODE
BETWEEN 50 DEG F AND 75 DEG F ([redacted])	ECONOMIZER MODE
50 DEG F OR LESS ([redacted])	H[redacted]TING MODE

6. IN THE [redacted]UPIED MODE OF FREEZESTAT (LOW [redacted]) MODE:
   - THE SUPPLY [redacted] AND EXHAUST [redacted]
   - THE ENERGY RECOVERY WHEEL SHALL BE OFF,
   - THE OUTSIDE [redacted] DAMPER AND EXHAUST [redacted] DAMPER SHALL BE FULLY CLOSED,
   - FACE AND BYPASS DAMPER SHALL BE IN FULL FACE POSITION.
7. WHEN PLACED INTO THE [redacted]UPIED MODE, [redacted] SEQUENTIAL ORDER AFTER OPERATION OF THE WATER-TO-WATER H[redacted]T PUMP SYSTEM HAS BEEN PROVED:
   - THE ENERGY RECOVERY WHEEL SHALL START AND OPERATION SHALL BE PROVED VIA CURRENT SWITCH,
   - THE OUTSIDE [redacted] DAMPER AND EXHAUST [redacted] DAMPER SHALL FULLY OPEN AND BE PROVED VIA CURRENT SWITCH,
   - THE SUPPLY [redacted] VFD AND EXHAUST [redacted] VFD SHALL START AND OPERATION SHALL BE PROVED VIA CURRENT SWITCH,
   - THE SYSTEM SHALL NOT START IF ANY ONE COMPONENT DOES NOT PROVE OPERATION, INCLU[redacted]NG THE HYDRONIC WATER-TO-WATER H[redacted]T EXCHANGER SYSTEM.
8. IN THE [redacted]UPIED MODE, THE FACE AND BYPASS DAMPERS SHALL MODULATE TO MAINTAIN [redacted]SCHARGE [redacted] TEMPERATURE ([redacted]) BASED UPON THE FOLLOWING SCHEDULE:

[redacted]SCHARGE [redacted] TEMPERATURE	MODE OF OPERATION
68 DEG F ([redacted])	COOLING MODE
VARIES	ECONOMIZER MODE
65 DEG F ([redacted])	H[redacted]TING MODE

(NOTE THAT THERE SHOULD BE NOT BE MORE THAN APPROXIMATELY 60% [redacted]FLOW NECESSARY ACROSS THE COOLING/H[redacted]TING COIL TO MAINTAIN REQUIRED [redacted]SCHARGE [redacted] TEMPERATURE. SET [redacted])

9. IF THE OUTSIDE [redacted] TEMPERATURE IS BETWEEN 60 DEG F ([redacted]) AND 65 DEG F ([redacted]), THEN THE ENERGY RECOVERY WHEEL SHALL BE OFF. THE WATER - TO -WATER H[redacted]T PUMP SYSTEM SHALL REMAIN OFF.
10. DEHUM[redacted]FICATION MODE: WHEN THE DUCT-MOUNTED EXHAUST [redacted] RELATIVE HUM[redacted]TY SENSOR R[redacted]DS 65% OR GR[redacted]TER, ENERGY RECOVERY WHEEL SHALL REMAIN ON AND BYPASS DAMPER SHALL CLOSE ALLOWING THE UNIT TO OPERATE WITH HALF [redacted]FLOW UNTIL EXHAUST [redacted] RELATIVE HUM[redacted]TY IS BELOW 60%.
11. [redacted] CONTROLLED BY THE VFD BASED ON FIELD-MOUNTED DUCT PRESSURE SETPOINT. [redacted] STATIC PRESSUREOPTIMIZATION SHALL BE UTILIZED BY POLLING OF ASSOCIATED [redacted]AV BOX DAMPER POSITIONS AND RESETTING OF PRESSURE SETPOINT AOALLOWED. DDC SYSTEM DETERMINES [redacted] BOX WITH GR[redacted]TEST DAMPER OPEN POSITION ONCE EVERY TEN MINUTES. UNIT'S SUPPLY [redacted] STATPRESSURE SETPOINT DECR[redacted]SED BY 0.1" WG AND 1.25" WG. SETPOINT IS 0.75" WG WHEN UNIT IS FIRST STARTED. DETERMINE IN THE FIELD WITH THE TAB CONTRCTOR THE STATIC PRESSURE SETPOINT TO OBTAIN ACCEPTABLE [redacted]FLOW AS DESIGNED.
12. HARDWARE INTERLOCK SUPPLY [redacted] EXHAUST [redacted] AND ISOLATION DAMPERS WITH EMERGENCY HVAC/VENTILATION KILL BUTTON.
13. THE EXHAUST [redacted] SHALL TRACK THE OUTSIDE [redacted] [redacted] V.A [redacted]FLOW STATION AND SHALL MAINTAIN A 1,000 [redacted] OFFSET FOR [redacted]CH UNIT TO MAINTAIN A POSITIVE BUIL[redacted]NG PRESSURIZATION. ALL RESTROOMS ARE PROVIDED WITH A CONSTANT VOLUME BOX AND SHALL MAINTAIN THE [redacted] IN[redacted]LATED ON THE DRAWINGS. IF WHILE THE EXHAUST [redacted] TRACKS THE OUTSIDE [redacted] THE [redacted] IN THE RESTROOM DROPS BELOW THIS [redacted]FLOW IN ANY ROOM BY 10%, THEN [redacted]
14. THE COIL IS A FULL FLOW COIL WITH FACE AND BYPASS DAMPER CONTROL FOR FREEZE PROTECTION AND DEHUM[redacted]FICATION CONTROL. LOW TEMPERATURE CUTOUT IS [redacted] LOCATED ON THE [redacted]SCHARGE OF THE COIL AND SHALL ACTIVATE WHEN THE L[redacted]VING [redacted] TEMPERATURE FALLS BELOW 35 DEG F. THE SUPPLY AND EXHAUST [redacted]S SHALL STOP, [redacted] OUTSIDE [redacted] AND EXHAUST [redacted] DAMPERS SHALL CLOSE.
15. LOW TEMPERATURE SENSOR [redacted] LOCATED ON THE [redacted]SCHARGE OF THE ENERGY RECOVERY WHEEL, WHEN THE L[redacted]VING [redacted] TEMPERATURE FALLS BELOW 40 DEG F THE SUPPLY AND EXHAUST [redacted]S SHALL STOP, OUTSIDE AND EXHAUST [redacted] DAMPERS SHALL CLOSE. WHEN TEMPERATURE RAISES TO 45 DEG F ([redacted]) SYSTEM SHALL RETURN TO NORMAL OPERATION.
16. A MANUAL RESET LOW [redacted] INSTALLED DOWNSTR[redacted]M OF THE CHILLED/HOT WATER COIL SHALL STOP THE OPERATION OF THE SYSTEM IF THE [redacted]SCHARGE TERMPATURE FALLS ' BELOW 35 DEG F.
17. A SMOKE DETECTOR SHALL BE LOCATED IN [redacted]CH [redacted] SYSTEM. IF SMOKE IS DETECTED, THEN THE SYSTEM SHALL SHUTOFF AND AN AL[redacted]OVISUAL ALARM SHALL ACTIVATE. UPON CORRECTION OF PROBLEM, THE SYSTEM SHALL BE RESET AND SHALL RETURN TO NORMAL OPERATION. COOR[redacted]NATE WITH [redacted] ANY [redacted] OF BUIL[redacted]NG'S [redacted] SHALL SHUTDOWN [redacted]
18. THE UNIT SHALL BE PROVIDED WITH A BUIL[redacted]NG WARM UP/COOL DOWN BYPASS DUCT. THIS BYPASS SHALL BYPASS THE WHEEL AND CR[redacted]TE A RECIRCULATING SYSTEM AND NOT USE ANY FRESH [redacted] SINCE THE BUIL[redacted]NG IS UN[redacted]UPIED. IF THE TEMPERATURE HAS FALLEN BELOW SETPONT THE [redacted]S SHALL BE ENGAGED AND THE UNIT SHALL [redacted]SCHARGE 95 DEG F ([redacted]). THE ASSOCIATED [redacted] BOX IN [redacted]CH ZONE SHALL OPEN USING THE OUTSIDE [redacted] UNIT TO BOOST TEMPERATURE. [redacted] IF THE TEMPERATURE RAISES ABOVE UN[redacted]UPIED SETPOINT, THE UNIT SHALL CONTROL TO A 55 DEG F ([redacted]) [redacted]SCHARGE [redacted] TEMPERATURE TO REDUCE THE UN[redacted]UPIED TEMPERATURE. THE ASSOCIATED [redacted] BOX IN [redacted]CH ZONE SHALL OPEN. ALL CONSTANT VOLUME TOILET EXHAUST BOXES SHALL BE CLOSED DURING WARMUP/COOL DOWN CYCLE.

OA-1 POINTS LIST

[redacted]SPLAY NAME	POINT NAME	BI	BO	BV	AI	AO	AV	ALARM TYPE	GRAPHIC
BUIL[redacted]NG STATIC PRESSURE	BLDG_SP				X			OUT OF RANGE	YES
DUAL TEMP MODULATING 2-WAY %OPEN	DT_CV-1_PO					X		[redacted]	YES
DUAL TEMP MODULATING 2-WAY OVERIDE	DT_CV-1_OVRD						X		
DEHUM[redacted]FICATION SETPOINT	OA_DHUM_SPT						X		YES
OA INTAKE HUM[redacted]TY SENSOR 1	OA_HUM_S_1				X				
[redacted]SCHARGE HUM[redacted]TY SENSOR 2	[redacted]_HUM_S_2				X				
OA [redacted] BYPASS DAMPER	OA_[redacted]_BYP_DAMP_C					X		[redacted]	
OA [redacted] BYPASS DAMPER STATUS	OA_[redacted]_BYP_DAMP_ST	X							
OA [redacted] BYPASS DAMPER OVERIDE	OA_[redacted]_BYP_DAMP_OVRD						X		YES
OA INTAKE DAMPER	OA_INT_DAMP_C					X		[redacted]	YES
OA INTAKE DAMPER STATUS	[redacted]	X							YES
OA INTAKE DAMPER OVERIDE	OA_INT_DAMP_OVRD						X		YES
[redacted] RELIEF DAMPER	[redacted]_INT_DAMP_C					X		[redacted]	YES
[redacted] RELIEF DAMPER STATUS	[redacted]_INT_DAMP_ST	X							YES
[redacted] RELIEF DAMPER OVERIDE	[redacted]_INT_DAMP_OVRD						X		YES
OUTSIDE [redacted] TEMPERATURE SENSOR 1	OA_TEMP_S_1				X				YES
OUTSIDE [redacted] TEMPERATURE SENSOR 2	OA_TEMP_S_2				X				YES
OUTSIDE [redacted] H/C TEMPERATURE SENSOR 3	OA_TEMP_S_3				X				YES
OUTSIDE [redacted] BYPASS TEMPERATURE SENSOR 4	OA_TEMP_S_4				X				YES
OUTSIDE [redacted] TEMPERATURE SENSOR 5	OA_TEMP_S_5				X				
OUTSIDE [redacted] AVERAGE TEMPERATURE	OA_TEMP_S_AVG_3&4						X		
OA PREFILTER STATUS	OA_FLTR_DP_S				X			OUT OF RANGE	
OA INTAKE HUM[redacted]TY SENSOR 1	OA_RH_S				X				YES
MODULATING FACE AND BYPASS DAMPER	OA_FBYP_DAMPER					X			YES
LOW [redacted] FREEZE STAT	LL-FRZ_STAT	X						OUT OF RANGE	YES
SCHEDULE	[redacted]_S			X					
OA [redacted]S START/STOP	OA[redacted]_RUN_C		X					[redacted]	YES
OA [redacted]S START/STOP OVERRIDE	OA[redacted]_RUN_OVRD			X				[redacted]	YES
OA [redacted]S STATUS	OA[redacted]_RUN_ST	X						[redacted]	YES
OA [redacted]S VFD FAULT	OA[redacted]_VFDFLT_S	X						BOOL[redacted]N CHANGE OF STATUS	YES
OA [redacted]S VFD SPEED	OA[redacted]_VFD_SPD_C					X			YES
OA [redacted]S VFD SPEED OVERRIDE	OA[redacted]_VFD_SPD_OVRD						X		YES
[redacted] 1 OA [redacted]	OA[redacted]1_AF_[redacted]				X				YES
[redacted] 2 OA [redacted]	OA[redacted]2_AF_[redacted]				X				YES
OA TOTAL [redacted]	OA_AF_[redacted]_TOTAL				X				YES
OA SMOKE DETECTOR	OA_SMOKE_DET	X						BOOL[redacted]N CHANGE OF STATUS	YES
OA STATIC PRESSURE	OA_SP_S				X				YES
OA STATIC PRESSURE SETPOINT	OA_SP_SETP						X		YES
OA STATIC PRESSURE RESET SETPOINT	OA_SP_RESET_SETP						X		YES
[redacted]UPIED COOLING SETPOINT	[redacted]_CLG_SETP						X		YES
[redacted]UPIED H[redacted]TING SETPOINT	[redacted]_HTG_SETP						X		YES
DUCT STATIC PRESSURE HIGH [redacted]	OA_DSP_HL	X						OUT OF RANGE	YES
DUCT STATIC PRESSURE LOW [redacted]	[redacted]_DSP_LL	X						OUT OF RANGE	
UN[redacted]UPIED BUIL[redacted]NG WARMUP H[redacted]TING SETPOINT	UN[redacted]_WARMUP_HTG_SETP						X		
[redacted] SMOKE DETECTOR	[redacted]_SMOKE_DET	X						[redacted]	
EXHAUST [redacted] TEMPERATURE SENSOR 6	[redacted]_TEMP_S_6				X				
EXHAUST [redacted] TEMPERATURE SENSOR 7	[redacted]_TEMP_S_6				X				YES
[redacted] PREFILTER STATUS	[redacted]_FLTR_DP_S				X			OUT OF RANGE	YES
ENERGY RECOVERY WHEEL [redacted]F PRESS. SENSOR	ERW_DP_S				X			OUT OF RANGE	
ENERGY RECOVERY WHEEL START/STOP	ERW_RUN_C		X					[redacted]	
[redacted] OVERIDE	ERW_RUN_OVRD			X				[redacted]	
[redacted]	ERW_RUN_ST	X						[redacted]	
[redacted] CONTROL	ERW_FRZ_C			X				[redacted]	
[redacted]S START/STOP	[redacted]_RUN_C							[redacted]	
[redacted]S START/STOP OVERRIDE	[redacted]_RUN_OVRD							[redacted]	
[redacted]S STATUS	[redacted]_RUN_ST							[redacted]	
[redacted]S VFD FAULT	[redacted]_VFDFLT_S							BOOL[redacted]N CHANGE OF STATUS	
[redacted]S VFD SPEED	[redacted]_VFD_SPD_C								
[redacted]S VFD SPEED OVERRIDE	[redacted]_VFD_SPD_OVRD								
[redacted] 1 [redacted]	[redacted]1_AF_[redacted]								
[redacted] 2 [redacted]	[redacted]2_AF_[redacted]								
[redacted] TOTAL [redacted]	[redacted]_AF_[redacted]_TOTAL								
SUM OF CONNECTED TERMINAL UNITS	TERM_UNIT_[redacted]_TOTAL				X				

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CONTROL SCHEMATICS

[redacted] OF BALTIMORE, MARYLAND

DATE	DESCRIPTION

M-8.0

03/13/2017
BID SET

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